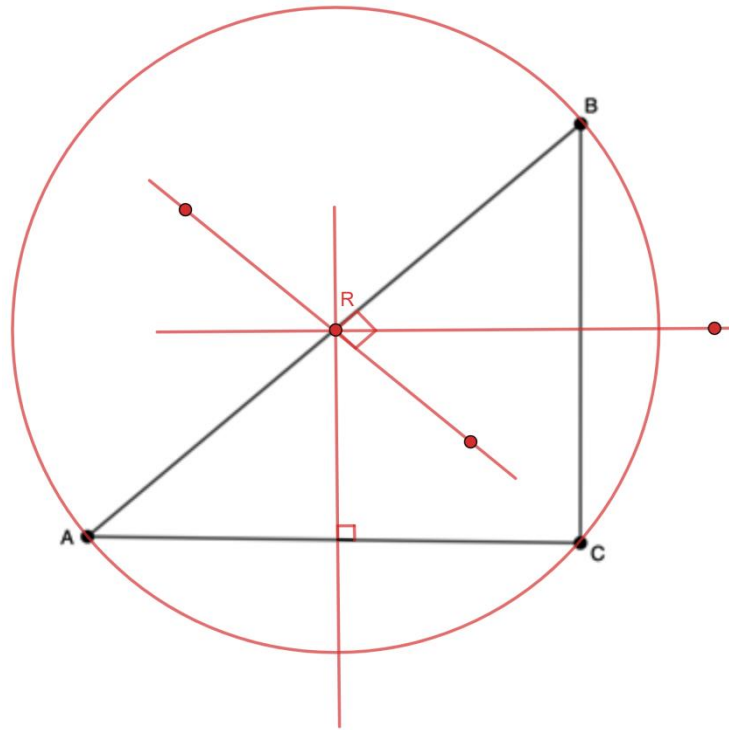


Geometry
Unit 11
Lesson 1 Practice

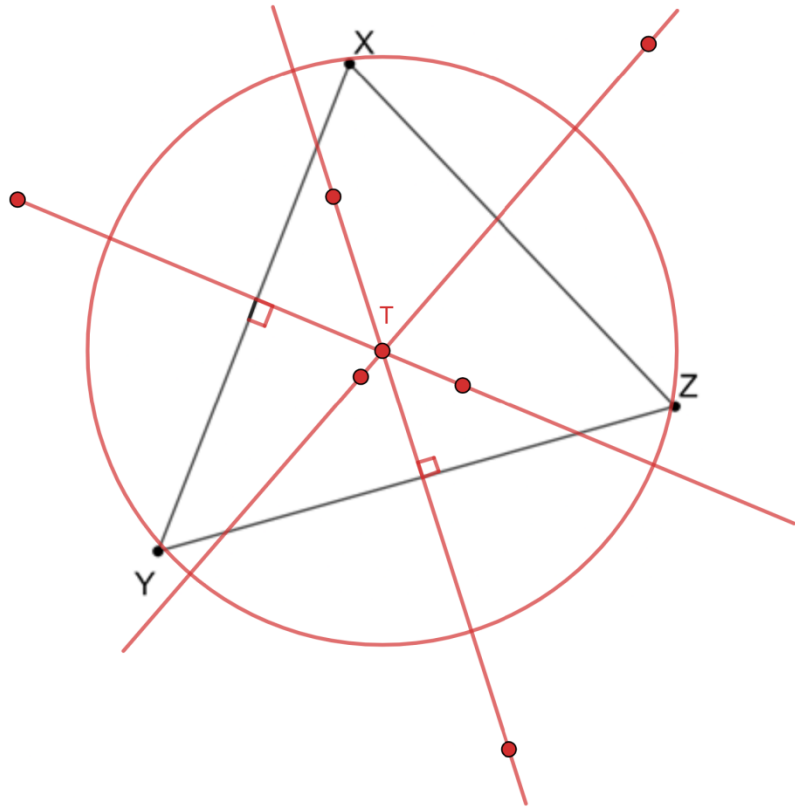
Name _____
Date _____
Period _____

1. Construct the three perpendicular bisectors for $\triangle ABC$.



2. Label the circumcenter of the triangle as Point R .
3. Use a ruler to measure the length of $\overline{AR}$.
1.5 in
4. Use a ruler to measure the length of $\overline{CR}$.
1.5 in
5. Use a ruler to measure the length of $\overline{BR}$.
1.5 in
6. Use a compass to draw the circumscribed circle around the triangle.

7. Construct the three perpendicular bisectors for $\triangle XYZ$.



8. Label the circumcenter of the triangle as Point T .

9. Use a compass to draw the circumscribed circle.

10. Given that the measure of $\widehat{XZ} = 100^\circ$, what is the measure of $\angle Y$?

$$m\angle Y = \frac{100}{2}$$

$$= 50^\circ$$

Geometry

Unit 11

Lesson 2 Practice

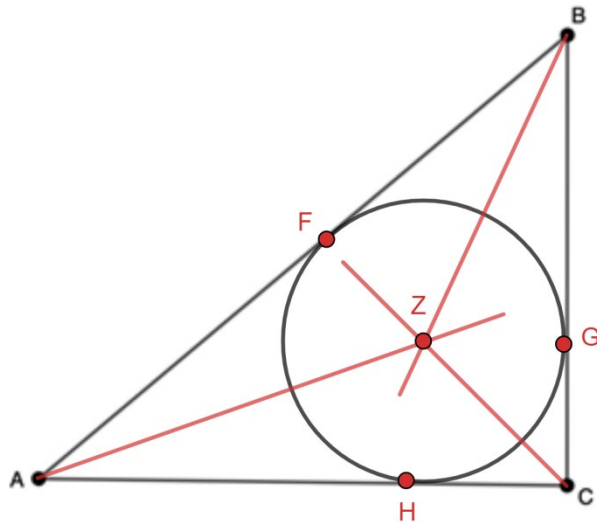
Name **Answer Key** _____

Date _____

Period _____

Use $\triangle ABC$ to answer questions 1-5.

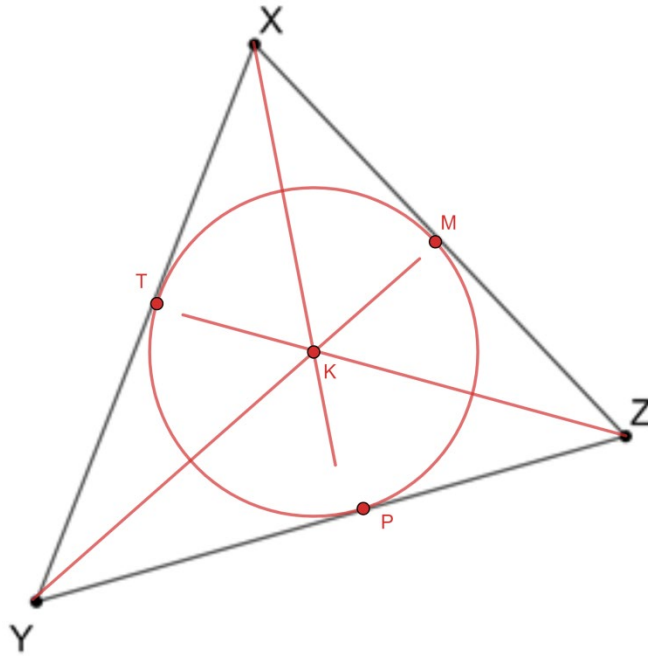
1. Construct the three angle bisectors for $\triangle ABC$.



2. Label the incenter of the triangle as Point Z.
3. Use a compass to draw the inscribed circle inside the triangle, with point Z as its center.
4. Label the three points where the inscribed circle intersects the triangle as F , G , and H .
5. Measure the lengths of segments $\overline{FZ}$, $\overline{GZ}$, and $\overline{HZ}$.
Answers may vary based on unit of measure, where $FZ = GZ = HZ$.

Use $\triangle XYZ$ to answer questions 6-10.

6. Construct the three angle bisectors for $\triangle XYZ$.

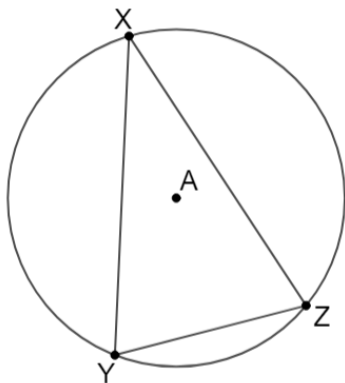


7. Label the incenter of the triangle as Point K .
8. Use a compass to draw the inscribed circle inside the triangle, with point K as its center.
9. Label the three points where the inscribed circle intersects the triangle as M , P , and T .
10. Measure the lengths of segments $\overline{MK}$, $\overline{PK}$, and $\overline{TK}$.
Answers may vary based on unit of measure, where $MK = PK = TK$.

Geometry
Unit 11
Lesson 3 Practice

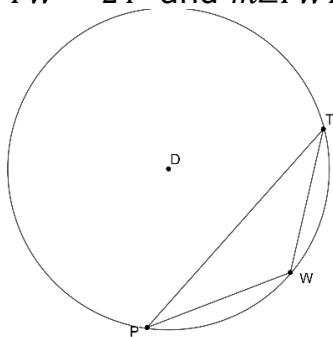
Name **Answer Key**
Date _____
Period _____

Circle A is shown, where points X, Y , and Z are on Circle A and create $\triangle XYZ$. $m\widehat{XZ} = 140^\circ$, $m\angle ZXY = 30^\circ$, and $m\angle XYZ = (2x - 18)^\circ$.



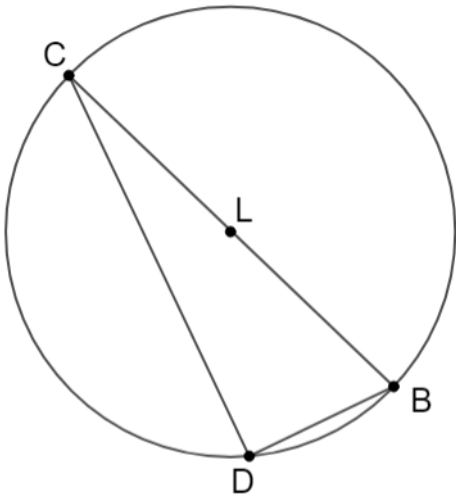
- Find the value of x . $2x - 18 = \frac{140}{2}$
 $2x = 88$
 $x = 44$
- Find the $m\angle XYZ$. $m\angle XYZ = 2(44) - 18 = 70^\circ$
- Find the $m\widehat{YZ}$. $2(30^\circ) = 60^\circ = m\widehat{YZ}$.
- Find $m\angle YZX$. $m\angle YZX + 30 + 70 = 180$
 $m\angle YZX = 80^\circ$

Circle D is shown, where points P, T , and W are on Circle D and create $\triangle PTW$. $m\angle PTW = 24^\circ$ and $m\angle TWP = 140^\circ$.



- Find $m\widehat{PT}$. $m\widehat{PT} = 2(140) = 280^\circ$
- Find $m\widehat{TW}$. $m\angle P = 180 - 140 - 24 = 16^\circ$; $m\widehat{TW} = 2(16^\circ) = 32^\circ$

Circle L has points B, C , and D on Circle L and diameter $\overline{BC}$. $\triangle BCD$ is inscribed in the circle. $m\widehat{CD} = 160^\circ$, $m\angle DCB = 10^\circ$, and $m\angle DBC = (2x + 20)^\circ$.

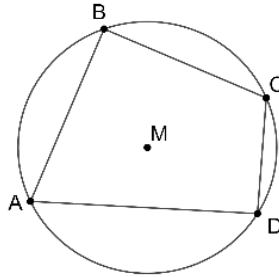


7. What is the $m\angle CDB$? $m\angle CDB = \frac{180}{2} = 90^\circ$
8. What is the value of x ? $2x + 20 = \frac{160}{2}$; $2x = 60$; $x = 30$
9. Determine $m\angle DBC$. $2x + 20 = 2(30) + 20 = 80 = m\angle DBC$
10. Determine $m\widehat{DBC}$. $m(\widehat{DBC}) = 360 - 160 = 200^\circ$

Geometry
Unit 11
Lesson 4 Practice

Name _____
Date _____
Period _____

Circle M is shown, with $ABCD$ inscribed in the circle. The $m\angle ABC = 96^\circ$, and $m\angle BCD = 125^\circ$.



1. Find the $m\angle ADC$.

$$180 - 96 = 84^\circ; m\angle ADC = 84^\circ$$

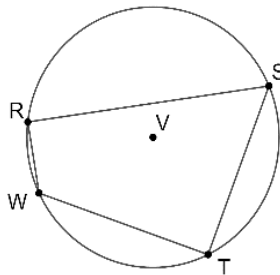
2. Find the $m\angle DAB$.

$$180 - 125 = 55^\circ; m\angle DAB = 55^\circ$$

3. Find the $m\widehat{ABC}$.

$$2 \cdot 84 = 168^\circ; m\widehat{ABC} = 168^\circ$$

Circle V is shown, with $RSTW$ inscribed in the circle. $m\angle RWT = 140^\circ$, $m\angle WRS = (5x)^\circ$ and $m\angle RST = (2x + 6)^\circ$.



4. Find the value of x .

$$(2x + 6) + 140 = 180$$

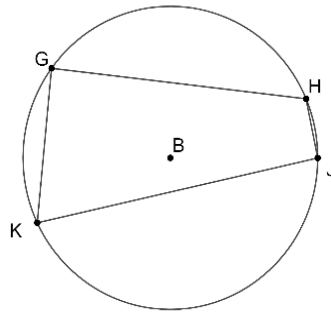
$$x = 17$$

5. Find the $m\widehat{WTS}$.

$$2 \cdot 5(17) = 170^\circ$$

$$m\widehat{WTS} = 170^\circ$$

Circle B is shown, with $GHJK$ inscribed in the circle. $m\angle GHJ = (3x + 50)^\circ$, $m\angle HJK = (4x)^\circ$, and $m\angle GKJ = (4x - 10)^\circ$.



6. Find the value of x .

$$(4x - 10) + (3x + 50) = 180$$
$$x = 20$$

7. Find $m\angle KJH$.

$$4 \cdot 20 = 80^\circ; m\angle KJH = 80^\circ$$

8. Find $m\angle GKJ$.

$$4 \cdot 20 - 10 = 70^\circ; m\angle GKJ = 70^\circ$$

9. Find $m\angle HGK$.

$$180 - 80 = 100^\circ; m\angle HGK = 100^\circ$$

10. Find $m\angle JHG$.

$$3 \cdot 20 + 50 = 110^\circ; m\angle JHG = 110^\circ$$

Geometry
Unit 11
Lesson 5 Practice

Name **Answer Key** _____
Date _____
Period _____

$\triangle MXB$ is shown where $m\angle M = (5x)^\circ$, $m\angle X = (15x - 10)^\circ$, and $m\angle B = (5x - 10)^\circ$.

1. Find the value of x .

$$\begin{aligned}5x + 15 - 10 + 5x - 10 &= 180 \\25x - 20 &= 180 \\x &= 8\end{aligned}$$

2. What is the $m\angle M$?

$$m\angle M = 5(8) = 40^\circ$$

3. What is the $m\angle X$?

$$m\angle X = 15(8) - 10 = 110^\circ$$

4. What is the $m\angle B$?

$$m\angle B = 5(8) - 10 = 30^\circ$$

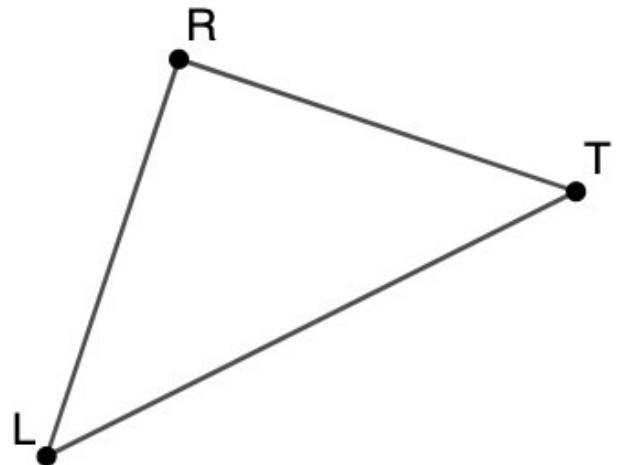
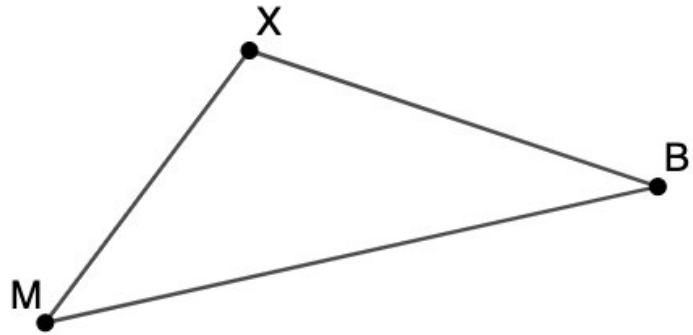
$\triangle LRT$ is shown, where $m\angle L = m\angle T = (2x + 6)^\circ$ and $m\angle R = (4x + 12)^\circ$. Figure is not drawn to scale.

5. Find the value of x .

$$\begin{aligned}2(2x + 6) + 4x + 12 &= 180 \\8x + 24 &= 180 \\x &= 19.5\end{aligned}$$

6. What is the $m\angle R$?

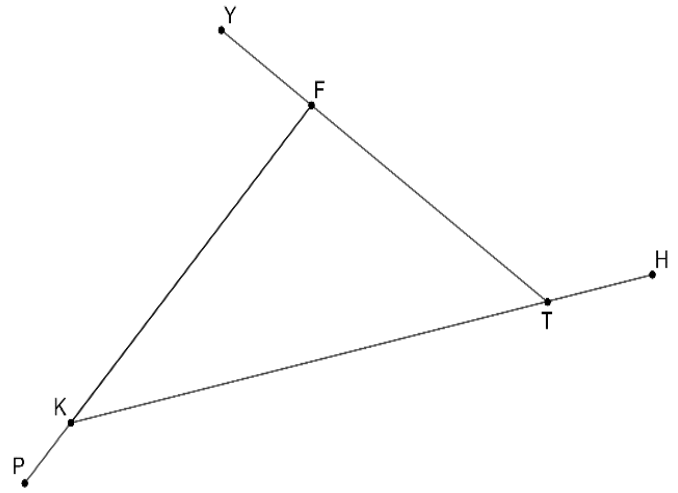
$$m\angle R = 4(19.5) + 12 = 90^\circ$$



$\triangle FKT$ is shown, where $\angle KFY$, $\angle PKT$, and $\angle FTH$ are exterior angles with $m\angle FTH = (12x + 25)^\circ$, $m\angle YFK = 95^\circ$ and $m\angle FKT = 24^\circ$.

7. Find the value of x .

$$\begin{aligned}12x + 25 + 95 + (180 - 24) &= 360 \\12x + 276 &= 360 \\12x &= 84 \\x &= 7\end{aligned}$$



8. What is $m\angle KTF$?

$$m\angle KTF = 180 - (12 \cdot 7 + 25) = 180 - 109 = 71^\circ$$

9. What is $m\angle PKT$?

$$m\angle PKT = 180 - 24 = 156^\circ$$

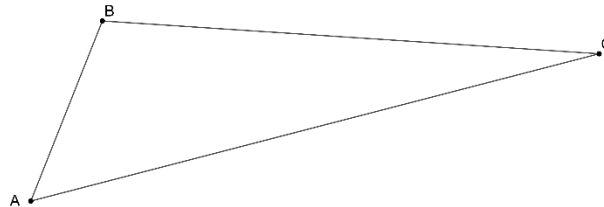
10. What is $m\angle KFT$?

$$m\angle KFT = 85^\circ$$

Geometry
Unit 11
Lesson 6 Practice

Name **Answer K**
Date _____
Period _____

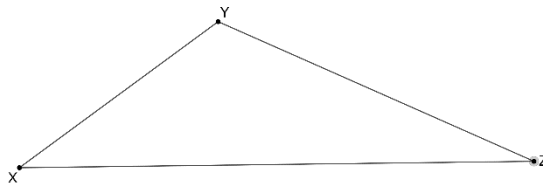
1. $\triangle ABC$ is shown, where $AB = 12$, $BC = 20$, and $AC = 26$.



List the angles in order from least to greatest.

$\angle C, \angle A, \angle B$

- $\triangle XYZ$ is shown. Use the triangle to complete questions 2 through 4.



2. Use a protractor to measure each of the angles.

$\angle X = 35^\circ$, $\angle Y = 120^\circ$, $\angle Z = 25^\circ$

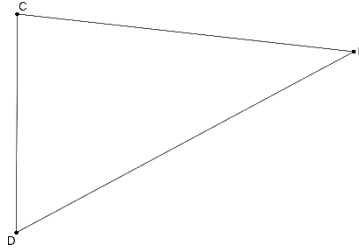
3. List the angles in order from smallest to largest.

$\angle Z, \angle X, \angle Y$

4. List the side lengths in order from smallest to largest.

$\overline{XY}, \overline{YZ}, \overline{XZ}$

5. $\triangle BCD$ is shown, where $m\angle B = 25^\circ$, $m\angle D = 70^\circ$, and $m\angle C = 85^\circ$.



List the side lengths in order from largest to smallest.

$\overline{BD}$, $\overline{CB}$, $\overline{CD}$

For questions 6 through 8, determine whether the given side lengths can form a triangle.

6. 10 cm, 11 cm, and 22 cm
 $10 + 22 > 11$; $11 + 22 > 10$; $10 + 11 < 22$
Not a triangle
7. 3 in, 7 in, and 14 in
 $3 + 7 < 14$; $3 + 14 > 7$; $7 + 14 > 3$
Not a triangle
8. 3 ft, 8 ft, and 6 ft
 $3 + 8 > 6$; $3 + 6 > 8$; $8 + 6 > 3$
Yes, forms a triangle

For questions 9 and 10, two sides of a triangle are given. Determine a possible side length for the third side.

9. 10 in and 2 in
Answers may vary. Value must be greater than 8 and less than 12.
10. 5 m and 3 m
Answers may vary. Value must be greater than 2 and less than 8.

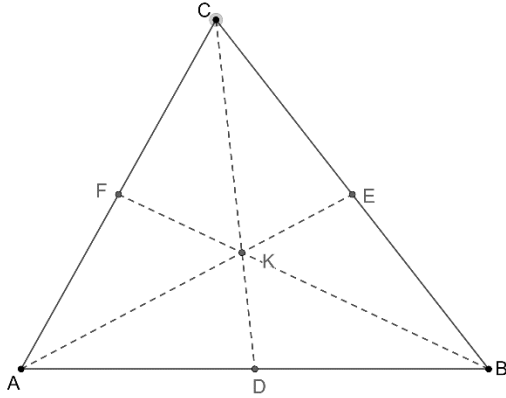
Geometry
Unit 11
Lesson 7 Practice

Name **Answer Key**
Date _____
Period _____

$\triangle ABC$ is shown. Complete the steps to for finding the centroid of the circle. Then, label the centroid Point G .

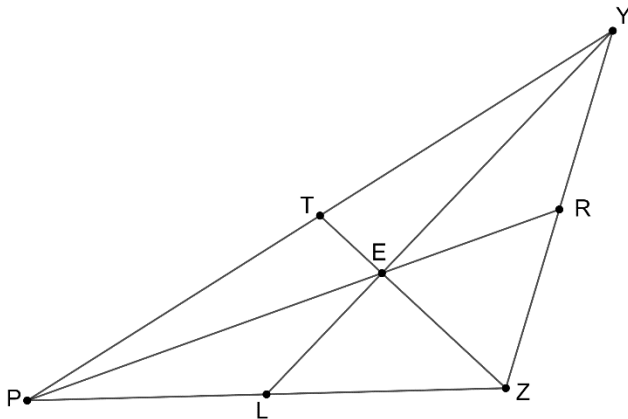
Steps	Construction
1. Construct the perpendicular bisector of $\overline{AB}$ to find its midpoint, R .	5.
Draw the median, R , to the opposite vertex, $\angle ACB$.	
2. Construct the perpendicular bisector of $\overline{BC}$ to find its midpoint, T .	
Draw the median, T , to the opposite vertex, $\angle BAC$.	
3. Construct the perpendicular bisector of $\overline{AC}$ to find its midpoint, S .	
4. Draw the median from Point S to the opposite vertex, $\angle ABC$.	
Draw Point G , the point of concurrency, where the three medians intersect.	

Triangle ABC is shown, where point K is the centroid of the triangle. $FB = 7.5$, $KC = 4.5$, and $KA = 4.8$. Use this information to answer questions 6 through 8.



6. Find the length of $\overline{CD}$. $\frac{2}{3}CD = 4.5$
 $CD = 6.75$
7. Find the length of $\overline{KE}$. $2KE = 4.8$
 $KE = 2.4$
8. Find the length of $\overline{BK}$. $\frac{2}{3}(7.5) = BK = 5$

Triangle YPZ is shown, where point E is the centroid of the triangle. $PE = 3x + 8$, and $RE = 2x - 4$. Use this information to answer questions 9 and 10.



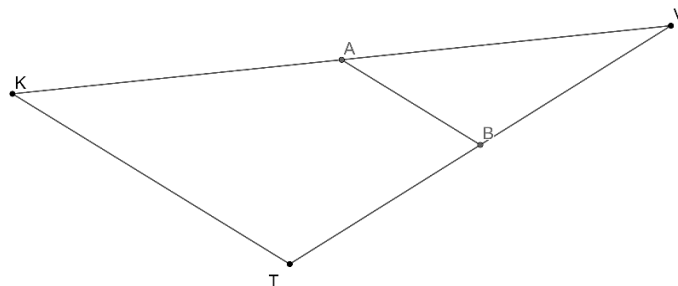
9. Find the value of x . $2(3x - 4) = 3x + 8$; $4x - 8 = 3x + 8$; $x = 16$
10. Find the length of $\overline{PR}$. $(3(16) + 8) + (2(16) - 4) = 84 = PR$

Geometry
Unit 11
Lesson 8 Practice

Name **Answer Key**
Date _____
Period _____

Use this information to answer questions 1 through 5.

Triangle KVT is shown, where $\overline{AB}$ is the midsegment. $KT = 2x + 4$, $AB = \frac{1}{3}x + 4$, $AV = \frac{3}{2}y + 3.5$, $KA = \frac{1}{2}y + 7.5$, $VT = 5z + 3$, and $TB = \frac{3}{4}x + 6$.



1. Find the value of x .

$$\begin{aligned}\frac{3}{2} \left[2 \left(\frac{1}{3}x + 4 \right) \right] &= 2x + 4 \\ x + 12 &= 3x + 6 \\ 2x &= 6 \\ x &= 3\end{aligned}$$

2. Find the value of y .

$$\begin{aligned}2 \left[\frac{1}{2}y + 7.5 \right] &= \frac{3}{2}y + 3.5 \\ y + 15 &= 3y + 7 \\ 2y &= 8 \\ y &= 4\end{aligned}$$

3. Find the value of z .

$$\begin{aligned}2 \left(\frac{3}{4}x + 6 \right) &= 5z + 3 \\ \frac{3}{2}(3) + 12 &= 5z + 3 \\ 5z &= \frac{27}{2} \\ z &= 2.7\end{aligned}$$

4. Find the length of the midsegment, $\overline{AB}$.

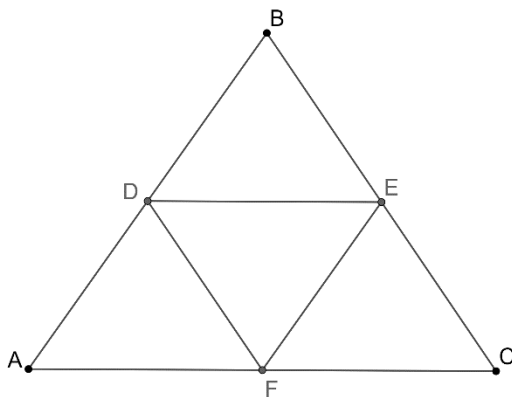
$$AB = \frac{1}{3}(3) + 4 = 5$$

5. Find the length of $\overline{KV}$.

$$\begin{aligned}KV &= KA + AV \\ KV &= \left(\frac{1}{2}(4) + 7.5 \right) + \left(\frac{3}{2}(4) + 3.5 \right) = 19\end{aligned}$$

Use this information to answer questions 6 through 10.

Triangle ABC is shown, where D, E , and F are midpoints. $DE = 2x - 4$, $AC = \frac{1}{2}x + 13$, $DF = \frac{3}{2}x$, $BC = 5x - 12$, $FE = 3x - 11$ and $AB = \frac{5}{3}x + 4$.



6. Find the value of x . $2(2x - 4) = \frac{1}{2}x + 13$

$$4x - 8 = \frac{1}{2}x + 13$$

$$8x - 16 = x + 26$$

$$7x = 42$$

$$x = 6$$

7. Find the length of $\overline{AB}$.

$$AB = \frac{5}{3}(6) + 4 = 14$$

8. Find the length of $\overline{BC}$.

$$BC = 5(6) - 12 = 18$$

9. Find the length of $\overline{AC}$.

$$AC = \frac{1}{2}(6) + 13 = 16$$

10. Find the perimeter of triangle DEF .

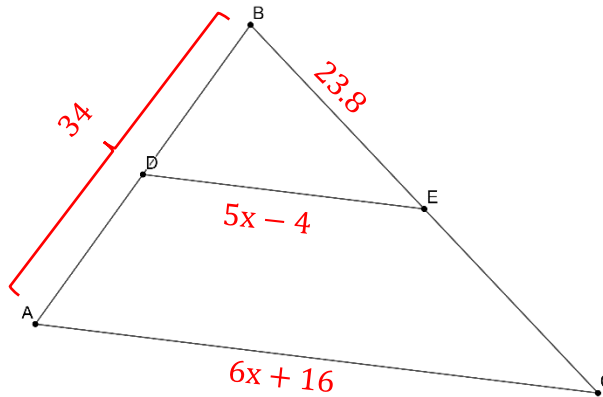
$$P = 14 + 18 + 16 = 48$$

Geometry
Unit 11
Lesson 9 Practice

Name **Answer Key**
Date _____
Period _____

Use this information to answer questions 1 through 5.

Triangle ABC is shown, where $\overline{DE}$ is the midsegment. $AC = 6x + 16$, $DE = 5x - 4$, $AB = 34$, and $BE = 23.8$



1. Find the value of x .

$$\begin{aligned} 2(5x - 4) &= 6x + 16 \\ 10x - 8 &= 6x + 16 \\ 4x &= 24 \\ x &= 6 \end{aligned}$$

2. Identify the length of $\overline{DE}$.

$$DE = 5(6) - 4 = 30 - 4 = 26$$

3. Identify the length of $\overline{AC}$.

$$AC = 6(6) + 16 = 52$$

4. Identify the length of $\overline{AD}$.

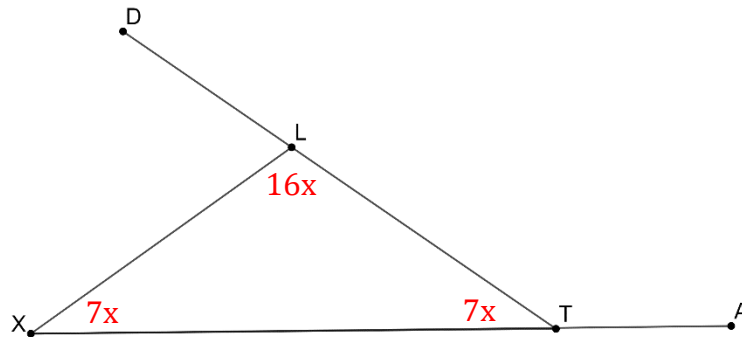
$$AD = \frac{34}{2} = 17$$

5. Calculate the perimeter of triangle BDE .

$$P = 17 + 23.8 + 26 = 66.8$$

Use this information to answer questions 6 through 10.

Triangle XTL is shown, where $\angle XLP$ and $\angle LTA$ are exterior angles. $m\angle LXT = m\angle LTX = (7x)^\circ$ and $m\angle XLT = (16x)^\circ$



6. Find the value of x .

$$\begin{aligned}7x + 7x + 16x &= 180 \\30x &= 180 \\x &= 6\end{aligned}$$

7. Find $m\angle XLT$.
 $m\angle XLT = 16(6) = 96^\circ$

8. Find $m\angle LTX$.
 $m\angle LTX = 7(6) = 42^\circ$

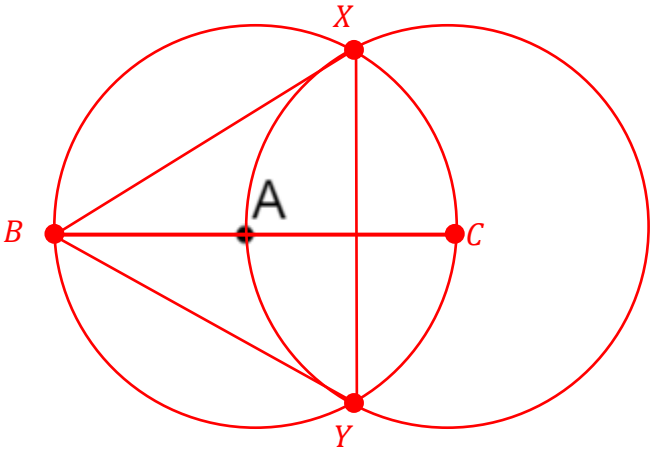
9. Find $m\angle DLX$.
 $m\angle DLX = 180 - 96 = 84^\circ$

10. Find $m\angle LTA$.
 $m\angle LTA = 180 - 42 = 138^\circ$

Complete the missing steps showing one way to construct an equilateral triangle inscribed inside a circle.

Step	Procedure
1	Use a compass to draw a circle. Mark the center and label it.
2	Draw a diameter. Label the endpoints B and C .
3	Without changing the span on the compass, place the compass on point C .
4	Draw a circle using point C as your center.
5	Label the points of intersection of the two circles circumferences as x and y (or any letter)
6	Draw segments in circle A connecting $\overline{BX}$, $\overline{XY}$, and $\overline{BY}$.

Use Point A to construct an equilateral triangle inside a circle.



Complete the missing steps showing one way to construct a square inscribed inside a circle.

Step	Procedure
1	Use a compass to draw a circle. Mark the center and label it.
2	Draw a diameter and label them M and K .
3	Construct a circle with a radius larger than the original circle with the center at one endpoint of the diameter.
4	Repeat step 3 using the other endpoint of the diameter.
5	Mark the intersections of the two circles, and label them T and R .
8	Connect the points, T and R with the endpoints of the diameter to create a square.

Use Point A to construct a square inside a circle.

